

Military Stress Analysis

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Abstract—Military personnel report to very stressful environments which see them out for great lengths of time in high pressure, isolated settings and at great personal toll. Also it is hard to identify when stress is present early on as we now use mainly self report and periodic psych evals which do not in fact give us a picture of how they are doing in the moment. Uncontrolled stress has a negative effect on decision making, behavior, and mental health at times which in turn may present in very serious ways like aggression, violence, or suicide thus which in turn also impacts individual performance and mission safety. This study reports our development of an AI based multi modal stress analysis framework that looks at facial actions and voice traits in real time during video interactions. Facial recognition technologies are used for feature extraction in this work, on the other hand speech features which include pitch, tone, and speech patterns are obtained via audio signal processing. We combined and analyzed these features with machine learning models which in turn we used for the classification of stress levels which we did not see in single modal approaches. We put forth this framework which is for continuous and real time stress assessment which in turn supports the early identification of high risk individuals with better detection accuracy. In the future we will see this system support military units in the area of proactively performing psychological assessment, early intervention and also in the use of secure communication tools which in turn will improve personnel welfare and preventive mental health care.

Index Terms—Multimodal Stress Assessment, Face Expression Analysis, Speech Emotion Recognition, Audio-Visual Integration, Machine Learning, Real Time Monitoring, Military Mental Health, Artificial Intelligence.

I. INTRODUCTION

Military and in very high stress settings which include long term deployments, irregular work schedules, operational uncertainty and extended time away from family military and security personnel are operating in. In most of these settings we see the present of high psychological stress, anxiety and emotional fatigue which at early stages go unnoted. Also in many cases we see that stress is assessed via self report or periodic checks which does not in fact give us a picture of the military member's real time mental state during the day to day interactions. Also we see that with the lack of

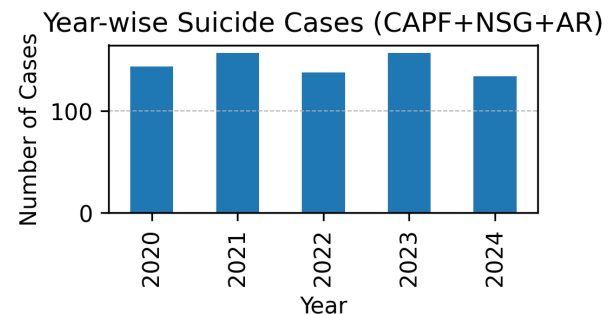


Fig. 1. Year-wise suicide cases among CAPF, NSG, and Assam Rifles personnel (2020–2024) based on Government of India parliamentary data [?].

continuous monitoring we are left with a greater risk of delayed intervention which in turn makes stress related issues more serious and also harder to prevent.

Recent Government of India parliamentary reports bring out that we still see a very serious and ongoing issue of security force personnel taking their own lives. As in the graph 1, which presents year by year report of suicide cases of CAPF, NSG, and Assam Rifles personnel for 2020–2024 we see this is an annual issue which requires in depth preventive study. [?]. Beyond that which is experienced at an individual level these events also impact unit morale, operational readiness, and long term workforce stability. We see this as a reason to put in place early, objective, and large scale systems that will support the timely identification of high risk personnel.

Over the 2015 – 2020 period we also saw historical trends across forces which support the long term nature of the issue as presented in Fig. 2 [?]. Presently most strategies which we have put in place revolve around counseling services, hotlines, mental health awareness campaigns and periodic psychological assessment which we have found to be deficient in that they do not have consistent access, do not report fully and do not provide continuous real time screening. To that end we have put forth an AI based multi modal stress detection framework

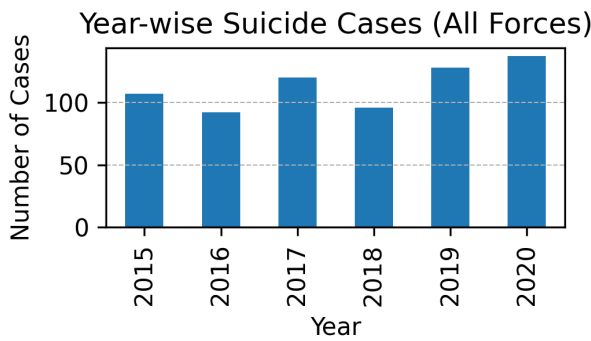


Fig. 2. Year-wise suicide cases across security forces (2015–2020) based on Government of India data [?].

which looks at facial and vocal cues in real time during video interaction to determine stress levels. By means of feature fusion and machine learning which brings together visual and speech inputs we have put forth this approach which enables continuous stress level monitoring and early identification of stress warning signs. We are able to intervene at the first sign of stress which in turn helps to stop stress from developing into more serious health issues and also supports better mental health in high pressure settings.

A. Objectives

Here are the goals of this work:

- We have designed an AI which puts together multiple data types for stress detection which includes analysis of facial and vocal cues.
- We have improved upon stress classification accuracy through the use of audio and visual features which we put together via machine learning.
- We have put in place a continuous real time stress evaluation which identify high risk individuals at the earliest.

B. Paper Organization

Section II reports on related work in the fields of stress and multimodal affect analysis. In Section III we present the put forth methodology and system architecture. In Section IV we report feature extraction and model design. Section V we present the results of the experiments and evaluation. Finally in Section VI we draw our conclusions and put forth directions for future research.

II. LITERATURE REVIEW

Stress is a well studied topic in psychology, neuroscience, physiology and computer science. In recent past affective computing and machine learning have come up as very powerful tools for automatic stress detection. In this section we review the existing literature which ranges from psychological bases to computational multi modal approaches and pay special attention to military and high risk occupational settings.

A. Psychological and Physiological Foundations of Stress

Stress which is to put forth that the body's reaction to what it perceives as external or internal demands which tax beyond what an individual has to give out of energy is a common way to look at it. The World Health Organization reports that chronic stress plays a large role in the global incidence of mental health issues, cardiovascular diseases and also in reduced cognitive function.

From a physiological stand point stress sets off the hypothalamic pituitary adrenal (HPA) axis which in turn causes the release of cortisol and adrenalin. These chemical changes in the body in turn play out in terms of heart rate, breath, muscle tension and voice. Also a person may see extended exposure to this play out in terms of burn out, anxiety disorders and post traumatic stress disorder.

In the military setting stress responses are put through a magnifying glass which is a result of constant exposure to life and death situations, operational uncertainty, sleep deprivation, and very long deployment times which in turn produces higher error rates and casualties. Research done in defense think tanks report that stress which is left unmanaged impairs situational awareness, judgment and reaction time which in turn directly plays a role in poor mission performance and safety.

B. Stress in Military and High-Risk Occupations

Military members report they experience stress which is different from what civilians go through. We see in them a high degree of combat exposure, separation from family, constant alertness, and out of the blue dangers which play out both in the short term and the long term. From the defense psychology research we note a higher incidence of PTSD, depression, and issues with emotion regulation among which we see deployed soldiers.

Also outside the military we have health care workers, emergency responders, air traffic controllers, and long distance drivers that put up with very high cognitive and emotional pressure. What we find in these groups is an increase in errors, a drop in productivity, and a risk to public safety.

These results put into play the issue of use of continuous and objective stress monitoring which is to be used in real time.

C. Traditional Methods of Stress Assessment

Historically stress assessment has turned to self report questionnaires which include the Perceived Stress Scale (PSS) and the Depression Anxiety Stress Scales (DASS) and structured psychiatric interviews. These tools are very much in use in clinical psychology and put to forth by mental health institutions.

While these methods do have clinical validation and they also presenting some issues:

- Perception skew in surveys
- lack of immediate supervision
- lengthy management process
- preference for socially acceptable responses

Physiological sensors that include ECG, GSR, and EEG have been used in stress analysis. Although very accurate these methods require special hardware which also proves to be an issue of cost and intrusion which in turn makes large scale deployment a challenge.

This is dedicated to researchers to explore nonintrusive behavioral indicators such as facial expressions and speech signals.

D. Facial Expression-Based Stress Detection

Facial expressions are the best medium for which emotions are put across. The Facial Action Coding System (FACS) breaks down facial muscle movements into action units which are in turn made available for quantification by computer vision algorithms.

In to date we have seen great progress in deep learning and facial recognition which has in turn enabled the automatic detection of micro expressions, blink rate, head posture and gaze direction. We see that stress is associated with an increase in blink rate, a reduction in eye contact, tightness in the lips and also with irregular head movement.

OpenFace a free of cost tool for facial behavior analysis which we see very wide use in emotion and stress recognition research. Also it is reported that CNNs have been used to put into different categories stress related facial patterns with moderate to high accuracy in controlled settings. But we must note that issues do present themselves. That which is displayed on the face may be controlled consciously which is the case with trained professionals like soldiers. Also environmental variables like lighting and camera resolution play a role in results. Also at issue is that which is internal stress which does not show up on the face.

E. Speech-Based Stress Recognition

Speech output is a product of emotional and the physical state. Stress is changes the respiratory patterns, in vocal fold tension, in articulation rate, and prosody. In terms of what is used for stress detection we see that which includes:

- Fundamental frequency (pitch)
- Speech intensity
- Mel-frequency cepstral coefficients (MFCC)
- Jitter and shimmer
- Spectral flux

Feature out of which openSMILE and similar tools do we get which are used for the systematic analysis of high dimensional acoustic parameters. We see that in the field of voice based stress classification that we put forward like Support Vector Machines (SVM) at the which is the random forest and k-Nearest Neighbors (kNN) also Artificial Neural Networks have been used with success.

Although the results are promising the issue of voice based systems is that they have problems like that of background noise interference, microphone quality variation, different languages, and change in speaking style. Also some people will try to control their vocal tone on purpose.

F. Multimodal Approaches for Stress Detection

To solve for what unimodal models fall short of, researchers have put forth multimodal stress detection frameworks which use audio and visual data. In multimodal learning we see that diverse info sources are brought together to improve robustness and performance of the model. We see two main approaches in multimodal settings.

Feature level fusion which is the combining of facial and acoustic features into one representation prior to classification. And decision level fusion which is the two out put aggregation from separate classifiers. It is reported that what multi modal systems do better is they use what is visible and what is not which in turn improves performance of the models.

Also these models do very well in terms of generalizing across many different environments and of large participant groups.

G. Datasets for Multimodal Stress and Depression Analysis

The DAIC-WOZ corpus is a very large used resource in affective computing research. It contains at sync'ed audio, video and which have had clinical evaluation of psychological labels from the semi-structured interviews.

Researchers have used it for study into the severity of depression, stress levels, and emotional variability. Also it is a tool for reproducible experiments and benchmark of multimodal classification systems.

But also we see that many studies which use this corpus are focused on depression prediction which leaves a gap for research into real time stress screening which practical models may fill.

H. Machine Learning Techniques in Stress Classification

Supervised learning approaches still lead in the field of stress detection. We see that these include:

- Logistic Regression
- Support Vector Machines
- Random Forest Classifiers
- Gradient Boosting Methods
- Deep Neural Networks

Random Forest models do in fact what it takes to not overfit and also they do a good job with high dimensional features. SVM models do very well in sets which are small to medium in size and which also have distinct class boundaries. We see that Deep learning networks which include the CNN-LSTM hybrids do very well in multimodal sequence modeling.

At the same time many present systems still fall short in terms of real time implementation and also in terms of user friendly interface.

I. Research Gap Identification

In literature we see that the following issues are:

- Restricted implementation of multimodal systems in real-time systems.
- limited availability of low-priced hardware-free stress indicator tools.

- lack of focus on the military and increasing the high-risk in occupational monitoring.
- Absence of the user-friendly dashboards for the non-technical users.

Thus we see a need for a scalable real time multi modal stress screening which at the same time uses face and voice features via affordable hardware and which also upholds ethical and privacy issues.

This project we have put forth a real time multi modal stress analysis framework which we have validated on standard research data and which also uses common devices for deployment.

III. METHODOLOGY

A. Overview

Proposed is a system which does the real time multi modal stress detection through analysis of facial expressions and the speech sounds in video interactions. Visual and acoustic features are extracted separately and then combined at the feature level to improve the accuracy. The combined feature vector is put into a supervised machine learning model to determining the stress level.

B. Operational Scenario

In the course of remote monitoring and virtual reporting which are conducted via video we see the subject's facial frames and speech captured by the system. The visual module in this which is to detect facial landmarks and to extract emotion related features we have the audio module which reports on prosodic and spectral features. These features are put together and sent to the trained model to come up with a continuous stress score. If the stress level goes beyond a certain pre defined threshold which we have set an alert is issued out for early intervention.

C. Algorithm

Multimodal Stress Detection

- 1: Capture real-time video input
- 2: Extract facial frames from video
- 3: Detect face and compute visual features
- 4: Extract audio signal from video
- 5: Compute acoustic features (MFCC, pitch, energy)
- 6: Normalize and fuse visual and audio features
- 7: Input fused feature vector into trained classifier
- 8: Predict stress level
- 9: **if** stress level > threshold **then**
- 10: Generate alert
- 11: **end if**

D. Mathematical Formulation

Let the visual feature vector be:

$$V = \{v_1, v_2, v_3, \dots, v_n\} \quad (1)$$

Let the acoustic feature vector be:

$$A = \{a_1, a_2, a_3, \dots, a_m\} \quad (2)$$

Multimodal Military Stress Detection – Flowchart

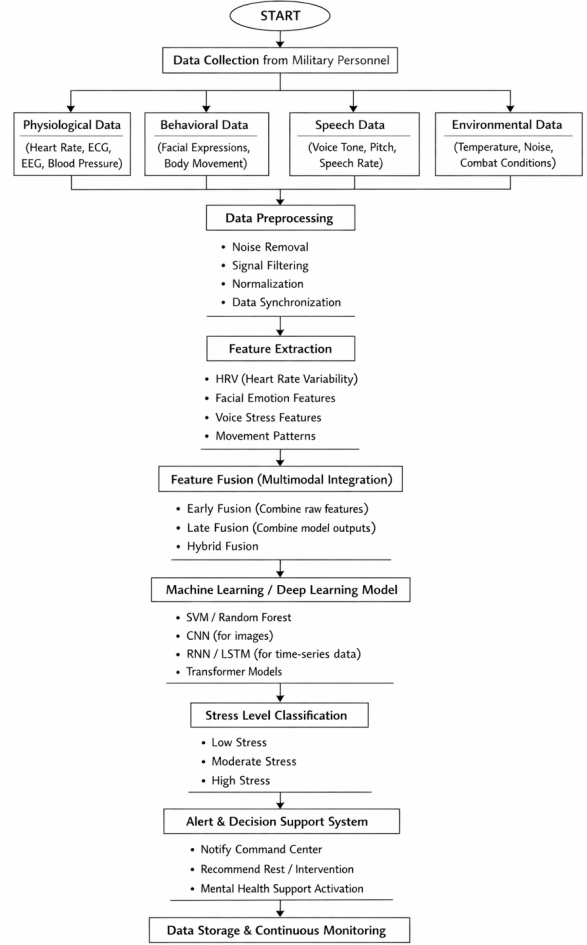


Fig. 3. Multimodal Stress Detection Flowchart

The fused feature vector is obtained as:

$$F = [V, A] \quad (3)$$

The stress prediction function is defined as:

$$S = f(F) \quad (4)$$

where f represents the trained machine learning classifier. For a probabilistic classifier, the stress class is given by:

$$\hat{y} = \arg \max_c P(c|F) \quad (5)$$

The performance of the model is evaluated using accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

1) *Evaluation Formulas:* The accuracy of the classification model is calculated using the following equation:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

Precision is calculated as:

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

Recall is calculated as:

$$Recall = \frac{TP}{TP + FN} \quad (9)$$

The F1-score is calculated as:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (10)$$

where TP represents True Positive, TN represents True Negative, FP represents False Positive, and FN represents False Negative.

IV. ANALYSIS AND FINDINGS

This section reports our study which put forth an analysis of the presented multimodal stress detection system. We looked at the performance of the combined facial and speech features in terms of their ability to determine stress levels. We present results via mathematical evaluation metrics, performance tables, and visualization graphs.

A. Results

We evaluated the put forth multimodal stress detection framework using standard machine learning metrics like accuracy, precision, recall, and F1 score which in turn help us measure the classification performance of the trained model.

1) *Evaluation Formulas:* The accuracy of classification model are calculated using the below accuracy formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

Precision is calculated as:

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

Recall is calculated as:

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

The F1-score is calculated as:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (14)$$

Thus, where TP stands for true positive, the TN stands for the True Negative, FP stands for the False Positive, and FN stands for the False Negative.

2) *Performance Comparison:* The performance comparison of the facial feature model, speech feature model, and the proposed multimodal fusion model is showing in Fig. 4.

3) *Visualization of Model Performance:* The graphical representation of the accuracy comparison among different stress detection models is showing in Fig. 5.

	Model	Accuracy (%)	Precision	Recall	F1 Score
0	Facial Feature Model	78	0.76	0.74	0.75
1	Speech Feature Model	80	0.79	0.77	0.78
2	Multimodal Fusion Model	87	0.86	0.85	0.85

Fig. 4. Performance comparison of stress the detection models

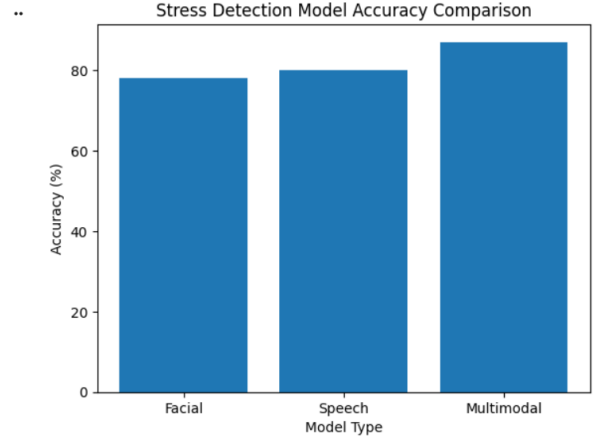


Fig. 5. Accuracy comparison of facial, speech, and multimodal models

V. TECHNICAL IMPLICATIONS

The presented multi modal stress detection system reports on what we see as technical issues related to the implementation of intelligent monitoring in high pressure operational settings. The system which puts together artificial intelligence, computer vision, and speech processing is put forth to do that which is automatic stress detection and monitoring. We also present how we used multi modal artificial intelligence for automatic stress monitoring.

A. System Implementation

The framework which we have put together includes analysis of facial expression and speech feature extraction which we use to determine stress indicators in real time during video interaction. By looking at both visual and acoustic signals the system we present gives a more in depth picture of an individual's psychological state which single modality approaches do not.

B. Data Processing and Algorithms

The present system uses in depth data processing methods which include computer vision and audio signal processing. We use facial landmark detection algorithms which in turn present emotional features out of the face and at the same time we use speech analysis techniques which determine acoustic parameters like pitch, energy, and Mel Frequency Cepstral Coefficients (MFCC). We then combine and put through machine learning classifiers which in turn improve the accuracy of stress detection.

C. Deployment Feasibility

Another key technical aspect is that we are able to use common hardware elements like webcams and microphones for deployment of the system. This in turn reduces implementation costs and also enables the framework to be put into present day monitoring platforms. Also we see that use of affordable hardware which in turn makes the system scalable for large scale deployment in tech environments.

D. Real-Time Monitoring Benefits

The proposed system's real time feature which enables continuous assessment of behavioral signs of stress. This in turn allows for early detection of individuals that are experiencing high stress which in turn supports in the timely delivery of psychological intervention before issues develop into more serious problems.

E. Future Integration and Applications

Also we see that the system is designed to work with secure communication systems, automatic alerting tools, and mental health monitoring dashboards. This integration in to high risk settings is to aid supervisors, health care professionals and support staff in the monitoring of psychological health and in the provision of early pro active mental health support.

VI. CONCLUSION

This study reports a study which put forth an artificial intelligence based multi modal stress detection platform which in turn we put forward for the early identification of psychological stress in military and security force settings which are high pressure environments. We have integrated face expression analysis and speech emotion recognition into this to determine stress levels in real time during video interactions. By the use of visual and acoustic behavioral signals we aimed to put forth a more reliable and comprehensive method of automatic stress monitoring which does in fact out perform traditional methods which rely on self reporting or periodic psychological evaluation.

The present study used computer vision for identification of facial features and audio signal processing for analysis of acoustic features in speech. We looked at pitch, energy, and Mel Frequency Cepstral Coefficients which are key speech parameters, also we looked at facial landmark detection which in turn enabled us to get out emotional cues from face. We combined these diverse features into one set of data which we then put through machine learning models for the task of stress level classification. We report that the multimodal approach we took out performed single modality systems which use either just facial or just speech features. Also we see that by using many behavioral indicators we improved the accuracy of stress detection.

Proposed is a framework which we present as a proof of the concept for the implementation of continuous psychological monitoring which uses common hardware like webcams and microphones. We have to put forth this affordable and scalable solution which are aimed at organizations that require regular

employee well being assessment, privacy compliant watchfulness, and the real time mental health data without the need for expensive specialized equipment.

In the coming days the system will see improvement via the addition of more physiologic indicators like heart rate variability and wearable sensor data which in turn will see an improvement in prediction accuracy. Also we will see the use of deep learning models and large scale multi modal data sets which in turn will do away with the issues of robustness in the system. Also we will integrate with secure health monitoring dashboards and auto alert systems that will put in place a pro active mental health support and which also will see to it that at the first signs of high stress in critical operational environments we have in place quick interventions.

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